MEENA E

Artificial Intelligence & Data Science | Software Engineering | Machine Learning
meenarekhadvelmrm@gmail.com | +91 8124336727 | Tiruvannamalai, Tamil Nadu
portfolio | github.com/meena | linkedin.com/in/meena-e

PROFESSIONAL SUMMARY

Artificial Intelligence and Data Science student at Madras Institute of Technology with a strong interest in AI, Machine Learning, and Software Engineering. Passionate about building scalable, data-driven solutions to solve real-world problems.

EDUCATION

B.Tech – Artificial Intelligence & Data Science — **CGPA: 8.58 / 10**
Madras Institute of Technology, Anna University, Chennai — *2023 – present*

Higher Secondary (HSC) — **Percentage: 96.8**
Centre of Academic Excellence, Saidapet, Chennai — *April 2023*

SKILLS

Programming	Python, C, C++, SQL
AI & ML	Machine Learning, Deep Learning, Computer Vision, Natural Language Processing,RAG
Libraries	TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV, Pandas, NumPy
Web Technologies	HTML, CSS, JavaScript, React.js, Node.js, Flask
Databases	MySQL, MongoDB
Core Concepts	Data Structures & Algorithms(100+ Leetcode problems), Object-Oriented Programming, Operating Systems, DBMS

PROJECTS

Intelligent RAG Framework for Legal Precedent Search & Q&A — ***Python | RAG | NLP | LLM | Semantic Search***

- Developed an evidence-grounded legal RAG system for semantic precedent retrieval and question answering from Indian legal judgments.
- Implemented hybrid sparse+dense retrieval using BM25 and SBERT/FAISS, followed by Reciprocal Rank Fusion (RRF) and authority-aware case reranking.
- Evaluated retrieval performance using MRR and Recall@10, achieving 0.95 MRR and 0.968 Recall@10.

Image Captioning using Deep Learning — ***Python | TensorFlow | Keras | CNN | LSTM | Computer Vision | NLP***

- Developed an image-to-text generation system using ResNet-50 for visual feature extraction and LSTM for sequential caption generation.
- Implemented an attention mechanism to dynamically focus on relevant image features while generating each word in the caption.
- Trained and evaluated the model on the Flickr8k dataset and deployed an interactive Streamlit application for real-time caption generation.

Text Dataset Analysis Dashboard — ***React.js | JavaScript | NLP | Data Visualization***

- Developed an interactive NLP analytics dashboard for uploading and analyzing text datasets through a React.js interface.
- Implemented token-level frequency analysis, word clouds, n-gram analysis, and sentiment distribution to extract insights from textual data.
- Built a dynamic CSV processing workflow with interactive visualizations for exploring dataset-level linguistic patterns.

Online Exam Management System — ***Python | PHP | Oracle Database | SQL | DBMS***

- Developed a database-driven web application for student registration, exam scheduling, question management, evaluation, and result processing.
- Designed and integrated relational database workflows using Oracle to manage examination-related records and application operations.
- Automated evaluation and result generation, reducing manual processing and improving the efficiency of examination workflows.

CERTIFICATIONS

- Big Data Computing – NPTEL (link)
- GenAI-Powered Data Analytics – Tata Forage (link)
- Generative AI – GUVI (link)
- SQL Developer – Infosys Springboard (link)

LANGUAGES

Tamil · English · Japanese (N5)